

Procta

AI-Proctored Exam Platform

Feature inventory · as of 28 May 2026

What's actually shipped — sourced from the codebase, not the roadmap. Every entry below corresponds to working code in production today.

1. Live Proctoring Engine

Multi-signal AI runs in the locked Electron browser on the student's machine. Frames stream to the teacher dashboard in real time over WebSocket; events stream over Server-Sent Events.

Face presence	Detects when the student leaves the camera. InsightFace embedding model (CPU). Logged as <i>face_missing</i> violation.
Wrong-person detection	Embedding compared against the ID-verification selfie at calibration. Triggers if a different person sits at the keyboard mid-exam. <i>wrong_person</i> violation, weighted heaviest in risk score.
Multi-face detection	Two faces in frame for >2 s → <i>multiple_faces</i> violation. Catches another person looking over the shoulder.
Gaze tracking	MediaPipe face-mesh + 9-point calibration grid. Logs sustained off-screen gaze. Per-student calibration quality tier (Tight / Loose / Suspicious) shown in Results.
Head-pose estimation	Yaw / pitch / roll from face landmarks; gating threshold tuned in calibration to per-student baseline (hardware-aware).
Cheat-object detection (YOLOv8 + SAHI)	Phone, headphones, earbuds (custom ear-crop classifier). Slicing-aided inference for objects partially out of frame. GPU auto-detect: CUDA → MPS → CPU.
Audio anomaly	Sustained voice / second-speaker detection via sounddevice RMS. Tuned to ignore typing and ambient noise.
Behavioural correlation engine	Multi-signal fusion: gaze + audio + face-pose patterns scored together to surface coordinated cheating (e.g. consistent gaze drift right + faint voice spikes).

2. Lockdown Browser (Electron)

The student's exam runs in a custom Electron build. The OS session is locked while the exam is active.

Tab-switch prevention	Browser focus loss for >1 s → <i>tab_switch</i> violation.
Process integrity scan	Live scan for VPN, remote-desktop, screen-share, debugger, AI assistant tools. Hyphen-aware regex (e.g. parsec-cloud no longer false-positives parsec).
Hardware-locked session	Camera + microphone permissions granted at calibration; lost permission mid-exam ends the session.
Auto-update via electron-updater	Hits GitHub Releases on launch. Existing installs upgrade on next relaunch with no manual intervention.

Offline answer resilience	Localstorage backup of every keystroke. On reconnect, background bulk-save catches up. Students never lose answers to a network blip.
Mac code-signed	Notarised .dmg distributed via /download/mac and /download/mac-x64. Windows code-signing pending Azure Trusted Signing.

3. Teacher Dashboard

Real-time control plane at app.procta.net. Single-page app with sticky chrome and single-scroll content per pattern.

Live sessions	47-student grid view with live face-frames over WebSocket, auto-updating risk scores, severity badges, force-submit, broadcast messages.
AI live-risk triage	One-line LLM summary per session, cached 60 s in Redis. Surfaces the 3 students most likely to be cheating right now.
Forensics timeline	Per-session scrub-through of every violation event with linked evidence screenshots. Downloadable PDF report.
Results table	Sortable, filterable, with per-row risk scores, calibration tier, and CSV / Excel / PDF / scorecard-ZIP exports.
Post-exam analytics	Score distribution (10 buckets), per-question difficulty + discrimination index, violation heatmap, risk band breakdown. Cached 60 s.
Real-time risk alerts	SSE toast notifications when a student's risk score crosses thresholds, with deep-link to the timeline.
Question editor + lint	MCQ / multi-select / true-false / short-answer types. AI lint runs each question pre-publish to surface ambiguity, leakage, or factual errors.
Question bank + import / export	Reusable pool of questions across exams. CSV / JSON bulk import with preview. Tag-based filtering.
Exam templates	Save an exam configuration + question set as a template; clone for next semester or section in one click.
Student groups / sections	Restrict exam access by group membership. Bulk add from registered students.
Live chat with students	Bidirectional WebSocket chat. Teacher broadcast or 1:1. Chat hub bounded with eviction (50 sockets/tenant, 4 h metadata TTL).
Onboarding wizard	5-step intro for first-time teachers. Persists <i>procta_onboarded</i> in localStorage; ? button re-opens.

4. Student Experience

Email invite → desktop install → calibration → exam. Calibration is mandatory and quality-scored.

Invite-driven enrolment	Teacher sends email invite; clicking the link auto-creates the student account on first /validate-student call. No pre-registration needed.
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9-point gaze calibration	Mandatory before exam start. Quality tier baked into the session record so teachers can flag suspicious-tight calibrations later.
Continuous identity verification	Face embedding sampled throughout calibration; mismatch aborts the exam with a typed <i>calibration_abort</i> event that pages the teacher.
ID-document verification	Selfie + photo-ID upload. Stored as evidence; teacher reviews in dashboard pending-verifications panel.
In-exam chat with invigilator	Floating action button on the exam window. Encrypted, auth-scoped to the session.
Practice mode	Sandbox session_id (PRACTICE_...) that exercises the full flow against canned questions. No DB writes.
Multi-exam lobby	Single auth → list of available exams → pick one → calibrate → write.
Student performance history	Longitudinal view across all exams the student has taken with this teacher (when student account is connected).

5. Grading and Assessment

Auto-graded MCQ / multi-select / T-F	Scored at submit. Multi-select uses set-equality. Shuffled-option mappings preserved per-session for accurate grading.
Short-answer with AI-suggested grading	Teacher writes reference + rubric + max_score. LLM proposes score + 1-2 sentence rationale + confidence (high/med/low). Teacher confirms or overrides; only <i>teacher_score</i> counts in the gradebook.
Rolled-up scoring	On grade-confirm, MCQ correct + sum(teacher_score) auto-recomputes the session total + percentage. Idempotent — safe to re-run.
Per-student scorecard PDF	Header, score summary, per-question table (student answer vs reference, correct/incorrect), violation summary, calibration tier.
Bulk scorecard ZIP	Streams scorecards for an entire exam as a ZIP via BytesIO. Doesn't load all PDFs into memory.
Email scorecards	Idempotent per <i>scorecard_emailed_at</i> . Teacher can click twice; already-emailed students skipped.
Composite risk score (0–100)	Log-saturating per-violation-type weights, duration-normalised. Cached. Recomputable via admin backfill endpoint.
AI session narrative	1-paragraph human-readable summary of a student's session (LLM over the last 80 violation events). Cached 60 s.

6. Operations and Infrastructure

FastAPI + async stack	All hot-path endpoints async via <i>_atable</i> ; supabase client wraps run in a worker thread for sync calls. 300+ concurrent students target on \$6 droplet.
Router decomposition	8 domain routers: auth / exam / admin / question_bank / grading / public / sse / chat. Single 7,000-line main.py shrunk to 382.
Pydantic strict-mode validation	ConfigDict(strict=True) on every request body model. Rejects type-coercion attacks (e.g. "123" for an int field) at the boundary.
Redis + in-memory caching	Two-layer: per-process LRU + Redis. Sorted-set LRU eviction for live frames (50-session cap). Cache-keyed risk scores, analytics, AI triage.
Atomic invite cap	Postgres RPC claim_invite_cap uses conditional UPDATE under row lock. Concurrent senders can't overshoot the daily cap.
Streaming CSV / PDF exports	StreamingResponse + async generators. Doesn't OOM on 5,000-row results tables.

Structured JSON logs + request IDs	Every request gets an X-Request-ID; logs are JSON for grep + log-aggregation tooling.
Sentry integration (optional)	SENTRY_DSN env var enables capture; off by default for local dev.
Security headers	CSP, HSTS, X-Frame-Options DENY, Permissions-Policy (camera + mic blocked at the app level for non-exam paths).
ETag + conditional requests	Middleware adds ETags to JSON responses ≤ 1 MB; 304 on matching If-None-Match.
GZip response compression	60-80% reduction on JSON >500 B. Static assets pre-gzipped at Docker build, served by Caddy.
Auto-update pipeline	GitHub Releases auto-discovered by backend (10 min TTL). Download buttons resolve to the latest release without redeploy.
Migration discipline	15 numbered SQL migrations, all idempotent. PGRST204 schema-cache drift handled with runtime fallback.
Test suite	14 pytest files: alerts, sessions, history, summary, behavioural analysis, e2e proctor, forensics timeline, invites, supporting modules. CI excludes heavy proctor tests via marker.

7. Communication and Workflow

Email invites with reminders	1 h + 24 h reminder cadence for unaccepted invites. Tracked via <code>opened_at</code> , <code>clicked_at</code> , <code>accepted_at</code> .
Real-time teacher↔student chat	Reconnection-tolerant WS. Per-session message history bounded by deque + cleanup loop.
Broadcast announcements	Teacher → all live students in an exam. Persists in chat history.
Force-submit + clear session	Time-windowed clear tokens (60 s TTL); audit-trailed violation event for every clear.
Webhook outbound (planned)	Schools' SIS systems. Stub exists; payload schema pending.

8. Privacy and Compliance Posture

RLS policies on all tables	Every table enforces <code>teacher_id</code> filtering at the DB level via Supabase RLS. Even if the API has a bug, cross-tenant data is unreachable.
JWT-scoped session ownership	Session keys embed <code>teacher_id</code> UUID; access validated against JWT <code>tid</code> claim on every authenticated call.
Encrypted at rest	Supabase Postgres encryption + DigitalOcean volume encryption.
HTTPS-only + HSTS	Caddy auto-provisions Let's Encrypt; HSTS preload-eligible.
Auto-delete face evidence (planned)	Phase-13 task: auto-purge face-detection screenshots at <code>exam_end</code> + 90 days for DPDP / FERPA / GDPR compliance.

9. Roadmap Snapshot

Items with working code or migrations queued. Prioritised by school-procurement leverage.

Windows code signing	Azure Trusted Signing \$10/mo. Removes SmartScreen warning for Windows downloads — currently deal-blocker for school IT.
Mobile PWA (lite proctoring)	Camera-based proctoring still works; tab-switch prevention doesn't. Position as low-stakes-quiz tool.
Inbuilt coding questions	Pyodide for Python in browser, LLM-judged for C++/Java. Phase 13a designed; build deferred per founder.
Per-criterion rubric grading	Extend short-answer to (clarity / correctness / depth) × marks. Universities specifically asked.

Cross-cohort plagiarism check	Cosine sim on embeddings of every short-answer pair. Cheap with a small embedding model.
Two-factor for teacher login	TOTP via authenticator app.
Bulk exam scheduling	Clone exam to N sections × M time slots. Coaching-institute ergonomic win.

10. Future Vision

Bigger bets — 6–24 months out. Not on the immediate sprint, but the direction Procta evolves to become a category-defining platform rather than a proctoring tool.

React dashboard rewrite	The current 7,000-line vanilla-JS dashboard is functional but ages every time a feature lands. Migrate to a React + Vite SPA matching the marketing site's design system. Wins: component reuse, dark/light theme switching, keyboard shortcuts, fluid loading states, drag-to-reorder, auto-escaping (kills the entire XSS class), and a real design language for sales screenshots.
Inbuilt coding questions	Pyodide for Python (browser-side WASM, zero server load). LLM-judged for C++/Java/Go where running untrusted code isn't worth the infra. Monaco editor with syntax highlight + autocomplete. Tests run on Run, hidden tests run on Submit. Phase-13a designed in Plan; deferred per founder.
AI exam generator from syllabus	Upload a PDF syllabus or paste a topic list; LLM produces a balanced 40-question exam tagged by Bloom's level + difficulty. Teacher tweaks → publish. Saves the 2-hour exam authoring session that's currently every teacher's worst job.
Voice-stream proctoring	Real-time speech-to-text on the audio feed. Flags when spoken keywords overlap with question text (collaborator reading aloud) or when prolonged conversational speech appears. Uses Whisper-small WASM for client-side transcription — no server-side audio storage.
Native mobile apps (iOS + Android)	React Native shell wrapping the same proctoring engine compiled for ARM. Position as the primary tool for low/medium-stakes assessments where the desktop install is friction. Camera-based proctoring still works; lockdown is weaker but compensated with stricter face-presence policy.
Teacher mobile app	Push-notify on high-risk events. Swipe-right to acknowledge, swipe-left to force-submit. For invigilators monitoring remotely. Pairs with the desktop dashboard, doesn't replace.
LMS integrations	Canvas, Moodle, Blackboard, Google Classroom plugins. Single-sign-on, exam launch from inside the LMS, scores auto-pushed back. The integration story is what unblocks university procurement deals.
Adaptive testing (IRT-based)	Questions adjust difficulty to the student's demonstrated level mid-exam, using item-response theory. Same total time, more accurate measurement, harder to brute-force-cheat (no two students see the same question sequence).

Teacher AI assistant	Chat with your gradebook in natural language: <i>“Which students improved most this semester?” “Show me question #14’s discrimination index across all sections.”</i> Backed by an LLM with read-only tool access to the teacher’s data.
AI proctor explainer for students	When a student is flagged, they see a plain-English explanation: <i>“Your gaze drifted off-screen for 12 s during Q14, which is longer than 95% of test-takers.”</i> Removes the “proctoring is a black box” hostility.
Cross-cohort plagiarism check	Cosine similarity on embeddings of every short-answer pair across the cohort. Highlights suspicious clusters. Cheap with a small embedding model run nightly.
Live invigilator marketplace	When AI flags a high-risk session, optionally route to a human reviewer (vetted, on demand, paid per minute). Schools that don’t have invigilator capacity buy review time from the platform. Two-sided revenue stream.
Multi-monitor detection	Browser screen-API + webcam reflection analysis to detect secondary monitors. Currently the loophole every prep-school student exploits.
rPPG attention + stress metrics	Heart-rate variability inferred from sub-pixel facial colour changes (research-grade rPPG). Surface attention dropoff and stress to teachers without storing biometric data — computed live, never persisted.
Whiteboard / stylus mode	Students show working on iPad / Wacom tablet for math + physics exams. Stroke-by-stroke recording for grading and review. OCR + LLM for partial credit on handwritten derivations.
Federated SSO for school districts	SAML / OIDC integration so a district admin manages 50 schools’ teachers from one console. Critical for K-12 district sales.
Public API + developer SDK	Third-party integrations on top of Procta — e.g. coaching institutes plugging Procta into their custom LMS. Rate-limited, scoped tokens, OpenAPI spec. Becomes a moat.
On-device LLM for free tier	Llama / Phi running locally for grading, lint, and risk narratives. Removes Groq dependency and cuts inference cost to zero for paying schools. Slower but private — useful selling point in EU.
Exam marketplace	Teachers publish question banks; others subscribe. Teachers earn revenue on usage. 30/70 split, Procta keeps 30%. Compounds the network effect once question-bank UX matures.

Generated from the Procta codebase. Sections 1–8 list features with shipping code or tracked migrations. Section 9 is near-term roadmap (working code or designs queued). Section 10 is directional — bigger bets that define where Procta goes next.